Lec.3: Microbial Metabolism

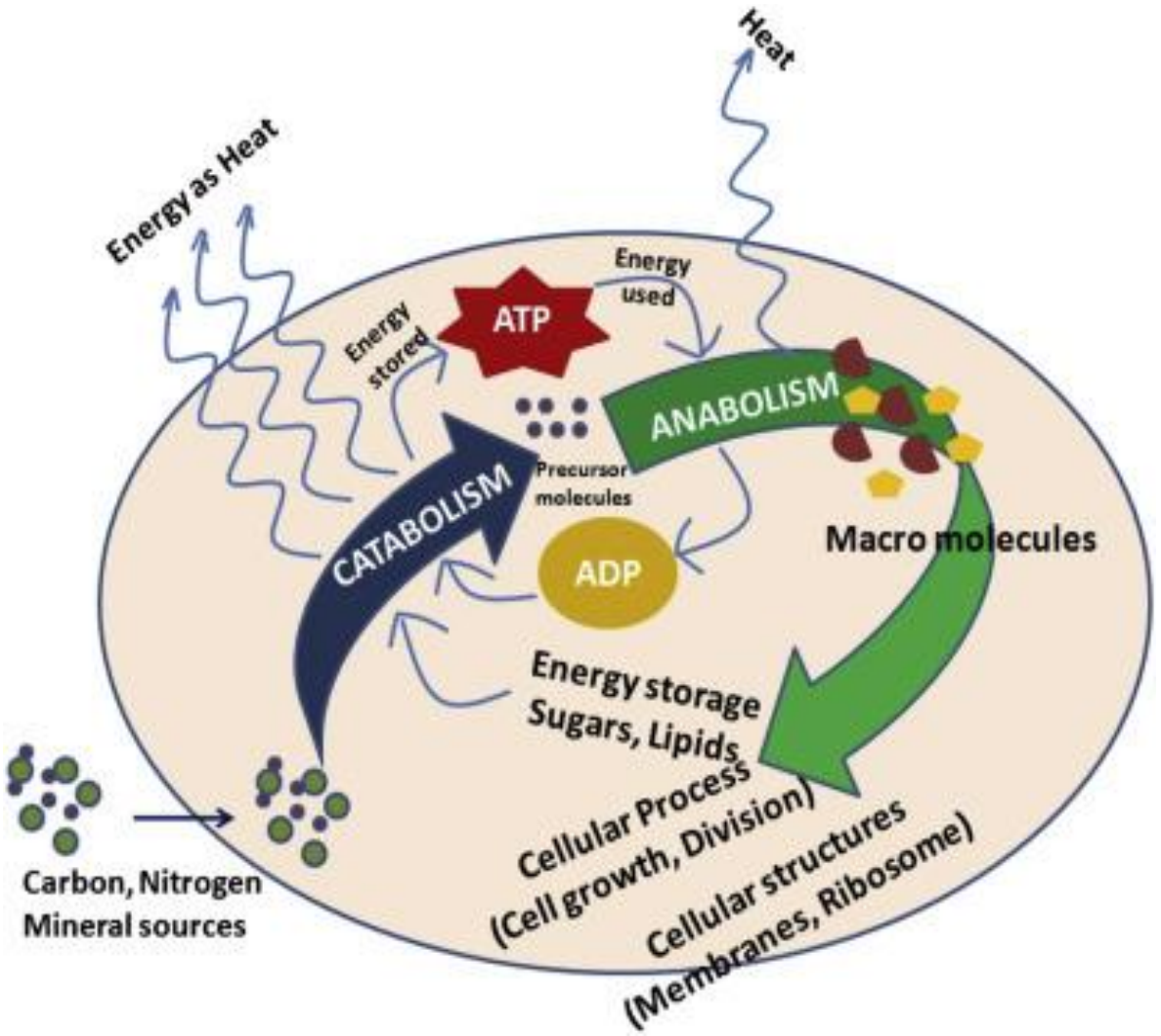
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Metabolism

- **Refer to the sum of all chemical reactions within a living organism. Because chemical reactions either release or require energy, metabolism can be viewed as an energy-balancing act.**
- **Accordingly, metabolism can be divided into two classes of chemical reactions: those that release energy and those that require energy.**

Catabolic and Anabolic Reactions

•In living cells, the enzyme is regulated that chemical reactions

- The crucial function of enzymes, therefore, is to speed up biochemical reactions at a temperature that is compatible with the normal functioning of the cell.

Although some enzymes consist of

1.Proteins component, called an apoenzyme.

2. cofactor or coenzyme

- Ions of iron, zinc, magnesium, or calcium
- Organic component : the B 1, 2 ,6 ,12 vitamin niacin or (nicotinic acid),
  - Nicotinamide adenine dinucleotide (NAD)
  - Nicotinamide adenine dinucleotide phosphate (NADP)and both function as electron carriers.
  - Flavin adenine dinucleotide (FAD),
- This coenzyme plays an important role in the synthesis and breakdown of fats and in a series of oxidizing reactions called the Krebs cycle.
- . Together, the apoenzyme and cofactor form a holoenzyme, or whole, active enzyme

Catabolism:

- The breakdown of complex organic compounds into simpler ones.
- Catabolic reactions are:
  1. Generally *hydrolytic reactions* (which use water and in which chemical bonds are broken), and
  2. they are *exergonic* (produce more energy than they consume).
  3. Catabolic reactions are coupled to ATP synthesis
- An example of catabolism occurs when cells break down sugars into carbon dioxide and water.

Anabolism:

the building of complex organic molecules from simpler ones.

Anabolic processes often involve

1. Generally *dehydration synthesis* reactions (reactions that release water), and
2. They are *endergonic* (consume more energy than they produce).
3. Anabolic reactions are coupled to ATP breakdown

Examples of anabolic processes are

1. the formation of proteins from amino acids,
2. nucleic acids from nucleotides
3. Polysaccharides from simple sugars.

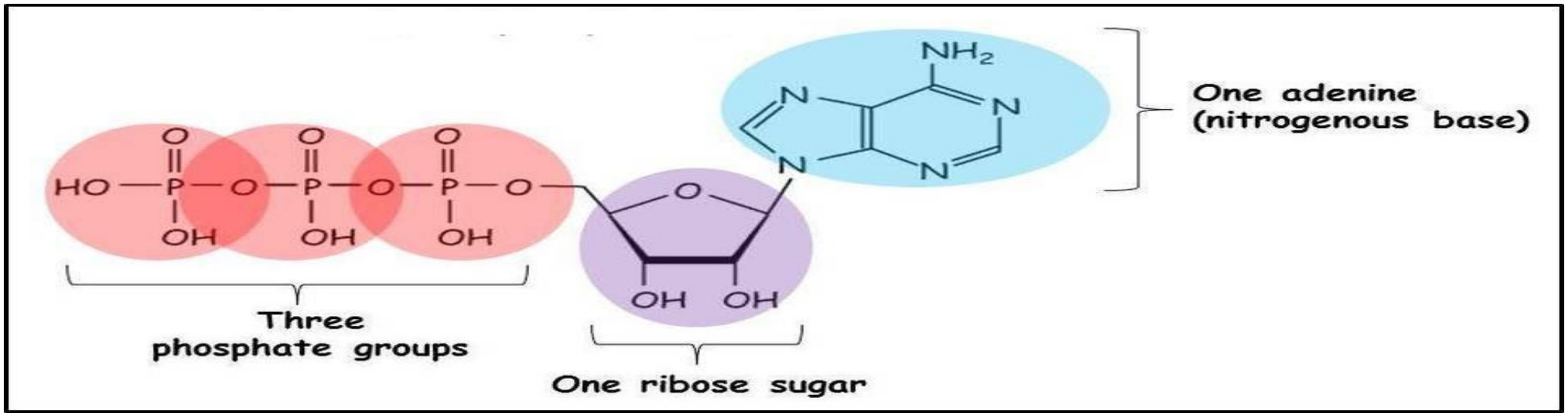
These biosynthetic reactions generate the materials for cell growth.

The structure of ATP

is a nucleoside triphosphate, consisting of

1. a nitrogenous base (adenine),
2. a ribose sugar,
3. and three serially bonded phosphate groups.

ATP is commonly referred to as the "energy currency" of the cell, as it provides readily releasable energy in the bond between the second and third phosphate groups.

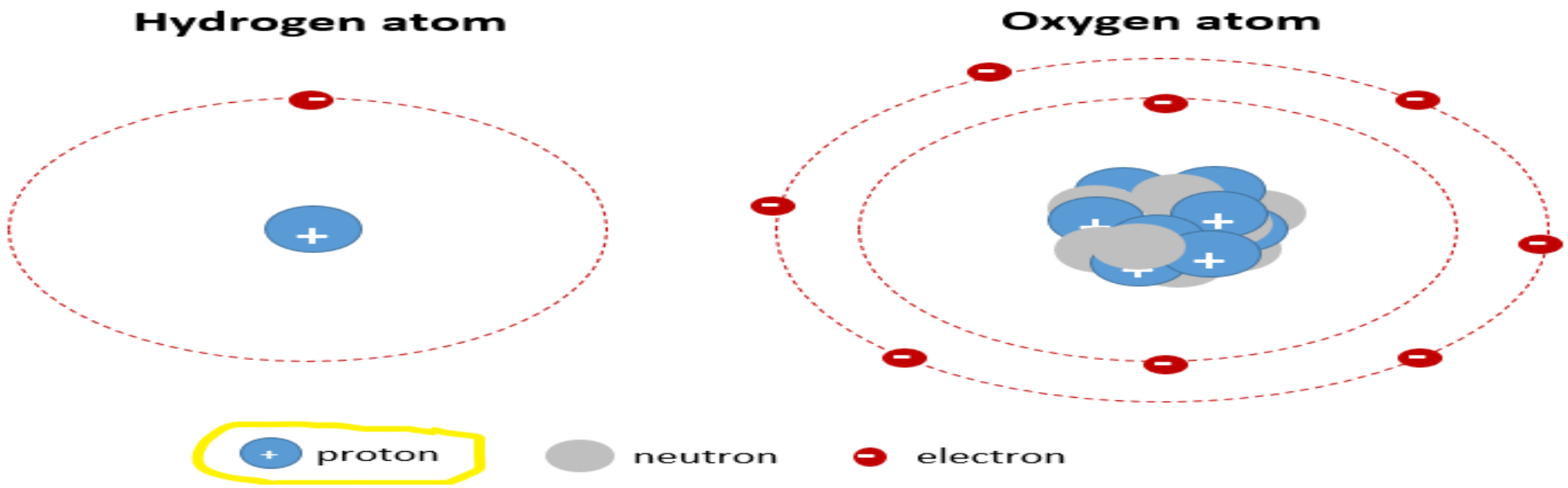


Organic compound, any of a large class of chemical compounds in which one or more atoms of carbon are covalently linked to atoms of other elements, most commonly hydrogen, oxygen, or nitrogen.

Oxidation is the removal of electrons (*e*) from an atom or molecule, a reaction that often produces energy.
Reduction (meaning that it has gained one or more electrons).*

Oxidation and reduction reactions are always coupled;

. The pairing of these reactions is called **oxidation-reduction** or a **redox reaction.**



The electron transport chain or Oxidative phosphorylation.

It includes the reactions that result in **the synthesis of ATP** from ADP + Pi.

- Where electrons are transferred from organic compounds to one group of electron carriers (usually to NAD and FAD). Then, the electrons are passed through a series of different electron carriers to molecules of oxygen (O_2) or other oxidized inorganic and organic molecules

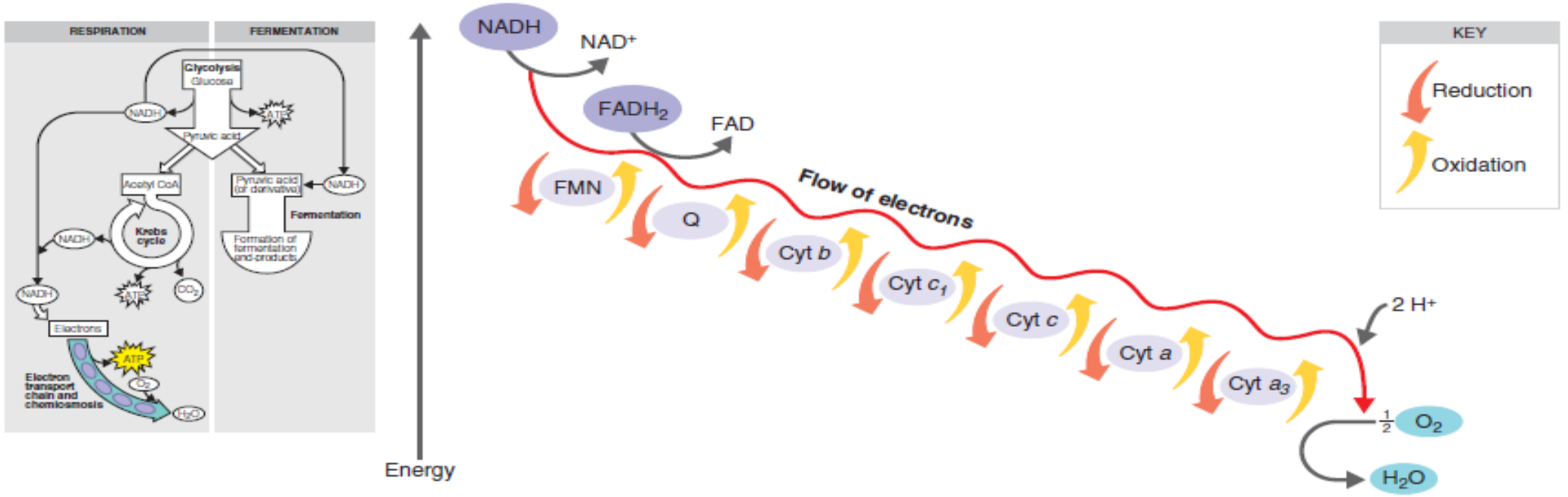


Figure 5.14 An electron transport chain (system). The inset indicates the relationship of the electron transport chain to the overall process of respiration. In the mitochondrial electron transport chain shown, the electrons pass along the chain in a gradual and stepwise fashion, so energy is released in manageable quantities. To learn where ATP is formed, see Figure 5.16.

- The reactions of the electron transport chain are carried out by a series of membrane proteins and organic molecules.
- They are arranged in four complexes.
- In eukaryotes, the electron transport chain is located in the inner **mitochondrial membrane**.
- In prokaryotes, it is located within the **plasma membrane**.
- Electrons move through the electron transport chain from a higher to lower energy state.
- Energy release moves protons through channels in the membrane proteins, moving them into the inner membrane space.
- This leads to a buildup of positively charged protons, which creates an electrical potential across the membrane.

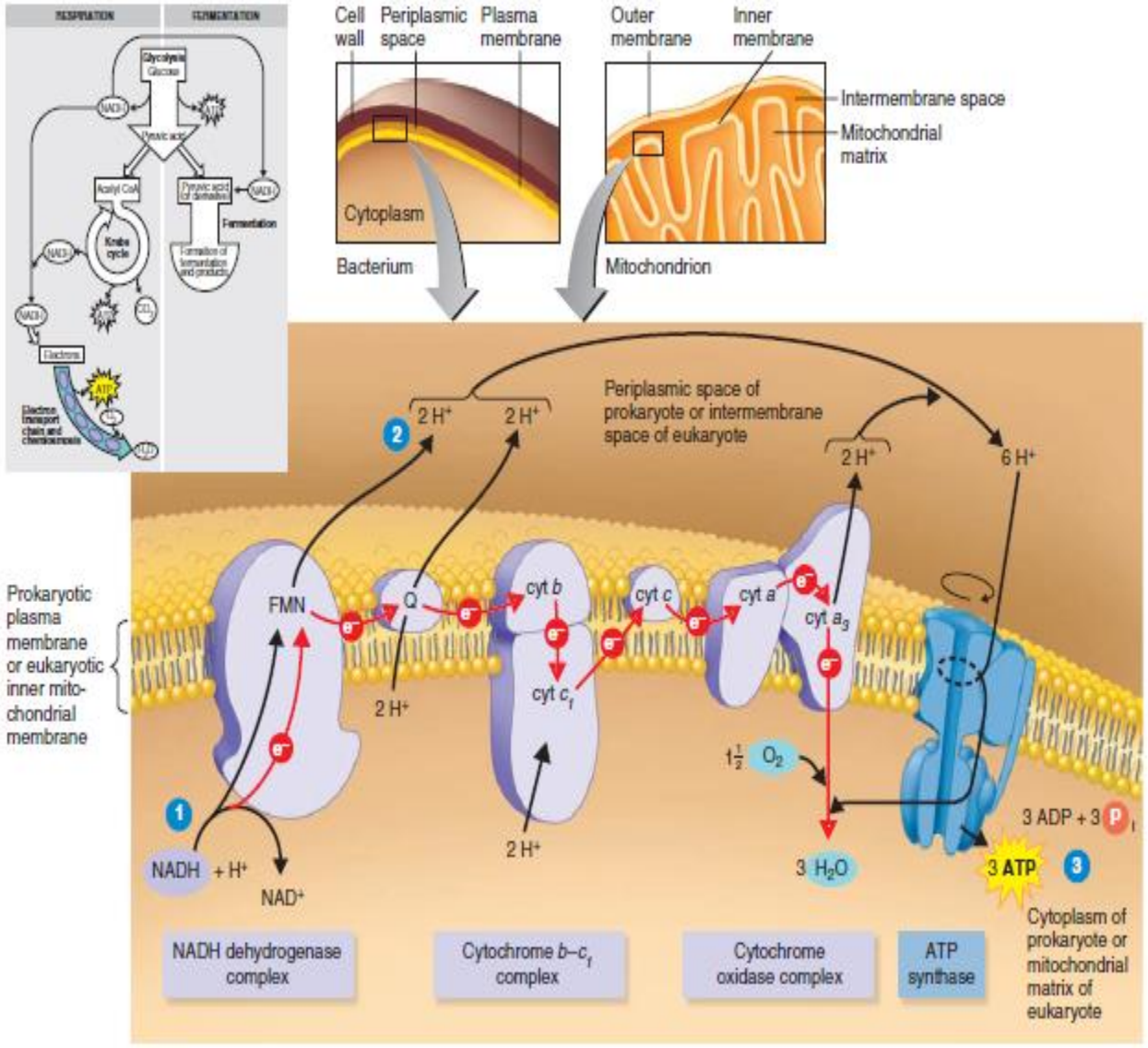


Figure 5.16 Electron transport and the chemiosmotic generation of ATP. Electron carriers are organized into three complexes, and protons (H^+) are pumped across the membrane at three points. In a prokaryotic cell, protons are pumped across the plasma membrane from the cytoplasmic side. In a eukaryotic cell, they are pumped from the matrix side of the mitochondrial membrane to the opposite side. The flow of electrons is indicated with red arrows.

Uptake of nutrients

Nutrient molecules frequently cannot cross selectively permeable plasma membranes through passive diffusion and must be transported by one of three major mechanisms involving the use of membrane carrier proteins.

Permeability absorption – Most microorganisms

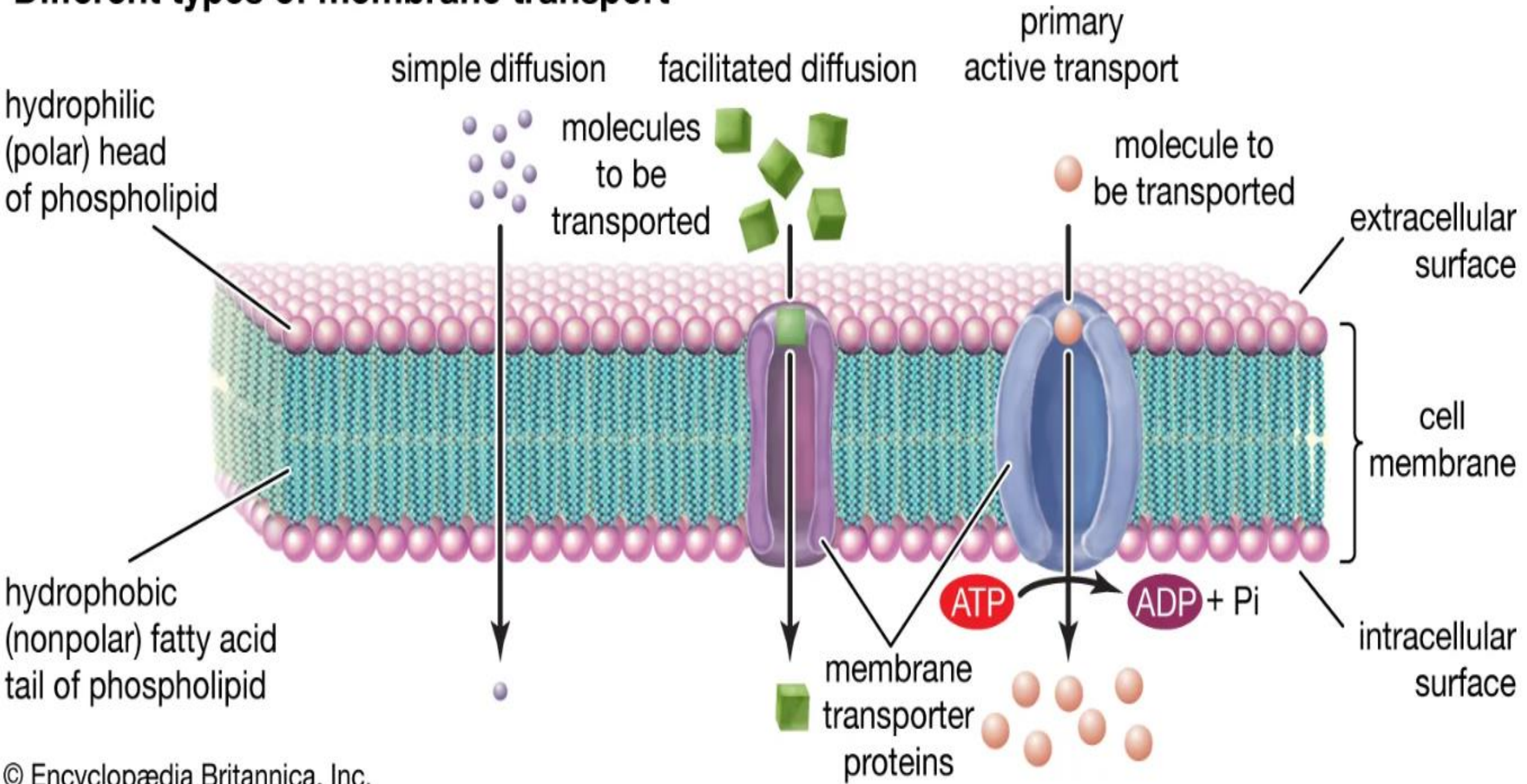
Passive Transport can be of the following different types:

- Simple diffusion
- Facilitated diffusion
- Osmosis

- Active Transport

- Group translocation

Different types of membrane transport



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1. Simple Diffusion

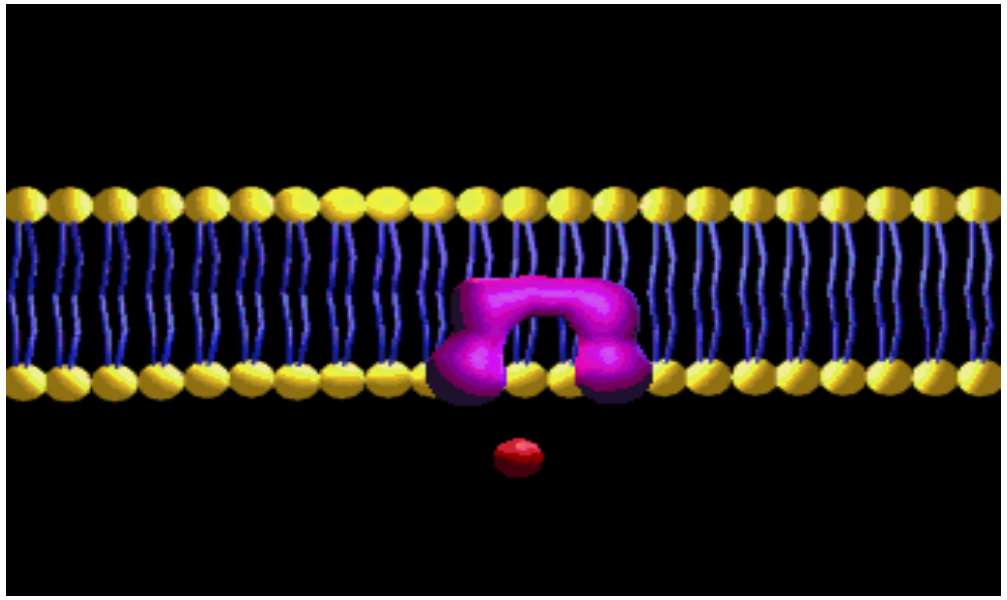
- A few molecules that are small can pass through the phospholipid bilayer by the process of diffusion Passive transport does not require an energy input
- Passive diffusion is the process in which molecules move from a region of higher concentration to one of lower concentration.
- The movement is down the concentration gradient

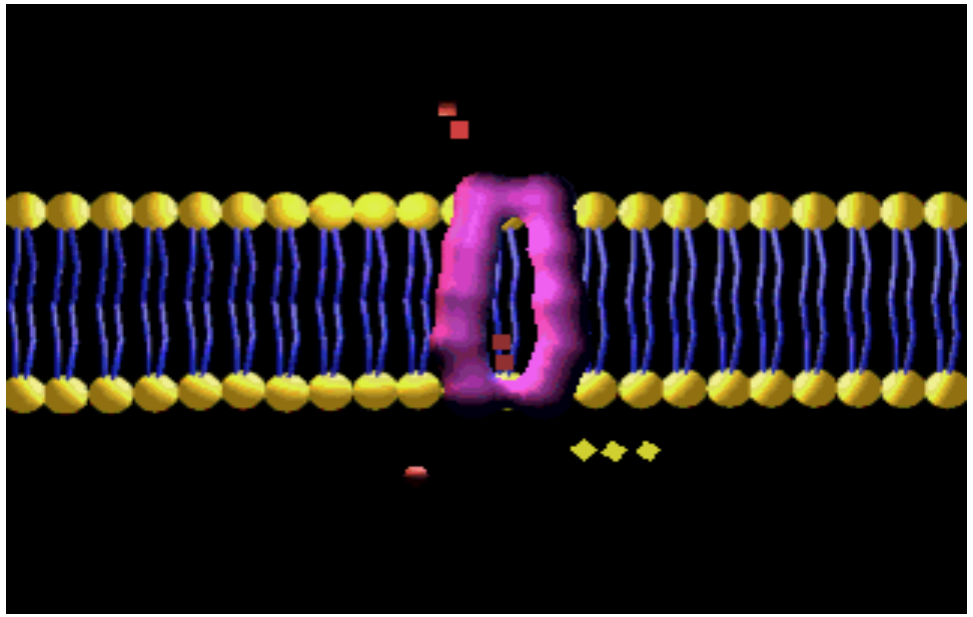
  no carrier protein;

  no energy.
- Slow process Based upon the free energy of the molecules themselves
- A few substances, such as **glycerol**, **H_2O**, **O_2** can cross the plasma membrane by **passive diffusion.**

2. Facilitated diffusion

- Higher con.→ lower con.
- Facilitated diffusion: carrier protein, called :permeases.
- **Permeases** are proteins which are embedded in the cell membrane
- They assist in the movement of other molecules across the membrane

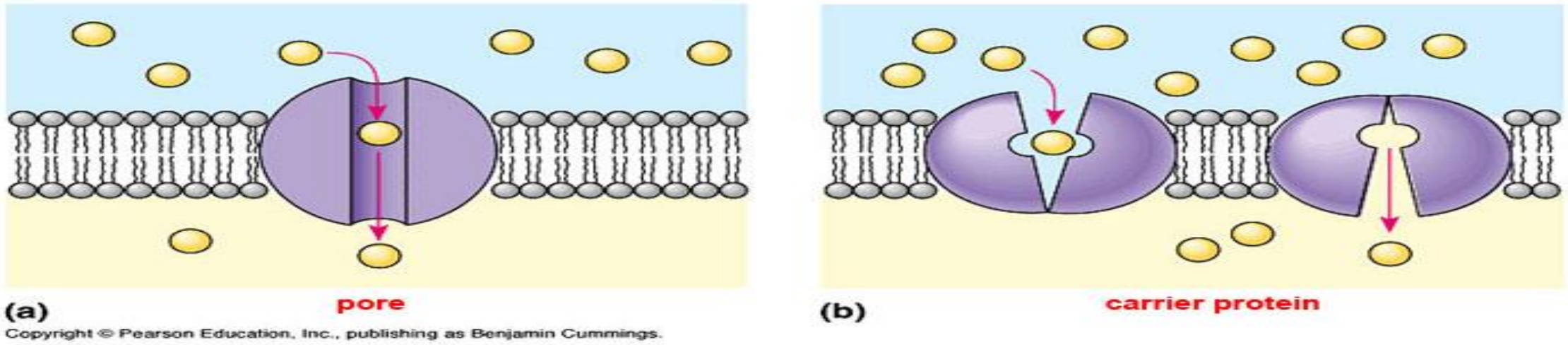




A model of facilitated diffusion

The membrane carrier can change conformation after binding an external molecule and subsequently release the molecule on the cell interior. It then returns to the outward oriented position and is ready to bind another solute molecule.

Because there is no energy input, molecules will continue to enter only as long as their concentration is greater on the outside.



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•3. Osmosis

•is the diffusion of water through a semipermeable membrane according to the concentration gradient of water across the membrane.

•Osmosis is the diffusion of water through a semipermeable membrane according to the concentration gradient of water across the membrane.

- A major permease in bacteria is **an aquaporin** that helps to move water in and out of the cell

Active transport

Active transport is the transport of solute molecules to higher concentrations, or against a concentration gradient, with the use of metabolic energy input.

•lower con.→ higer con.

•Permeases, energy

- Transport of molecules and ions against the concentration gradient from low concentration to high concentration
- Requires the input of energy in the form of ATP
- ATP = ADP + P + energy

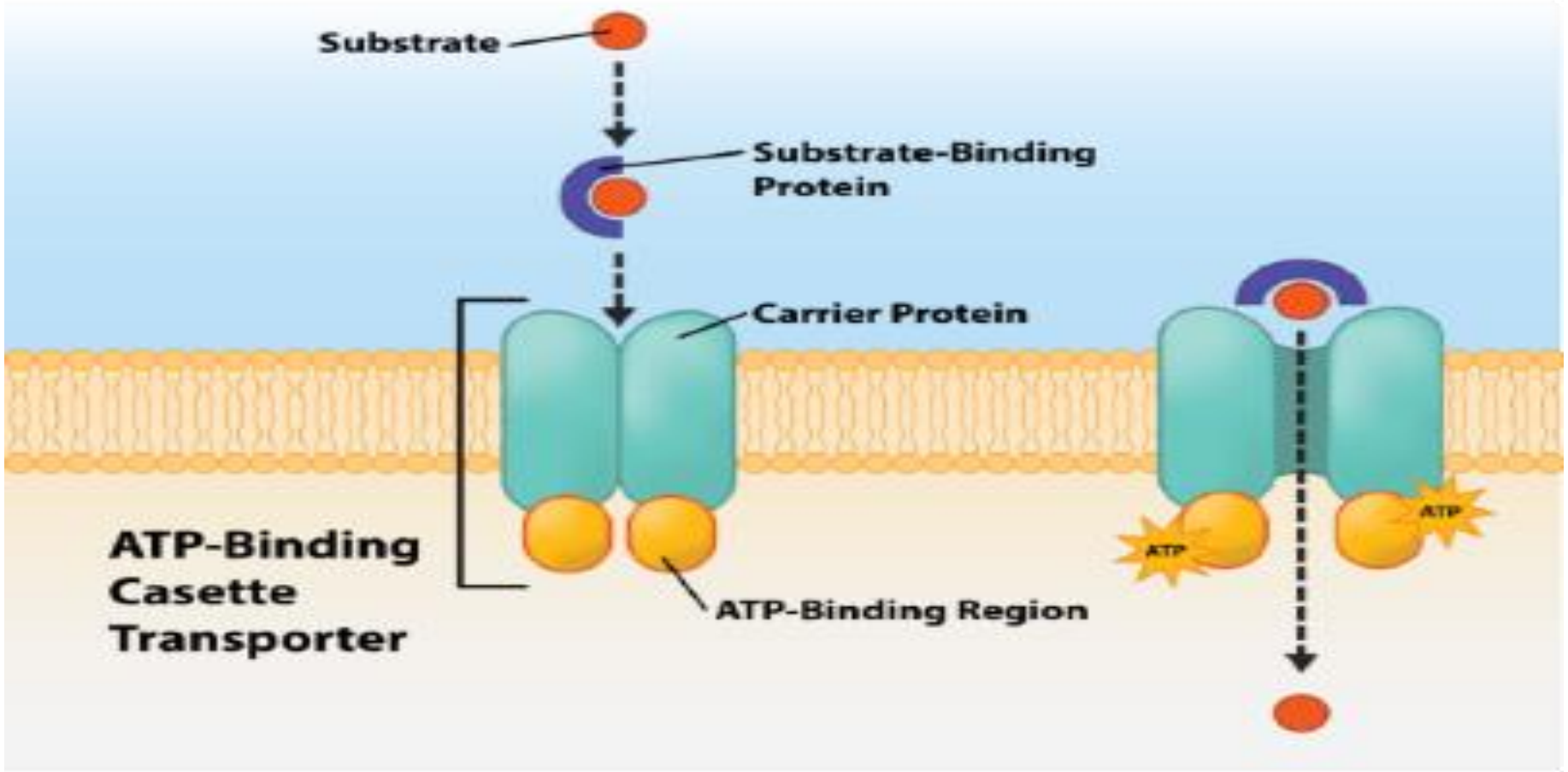
Group translocation

A biological process where a molecule crossing the cell membrane not only gets transported but also gets transformed in itself.

This modification requires the input of ATP energy .

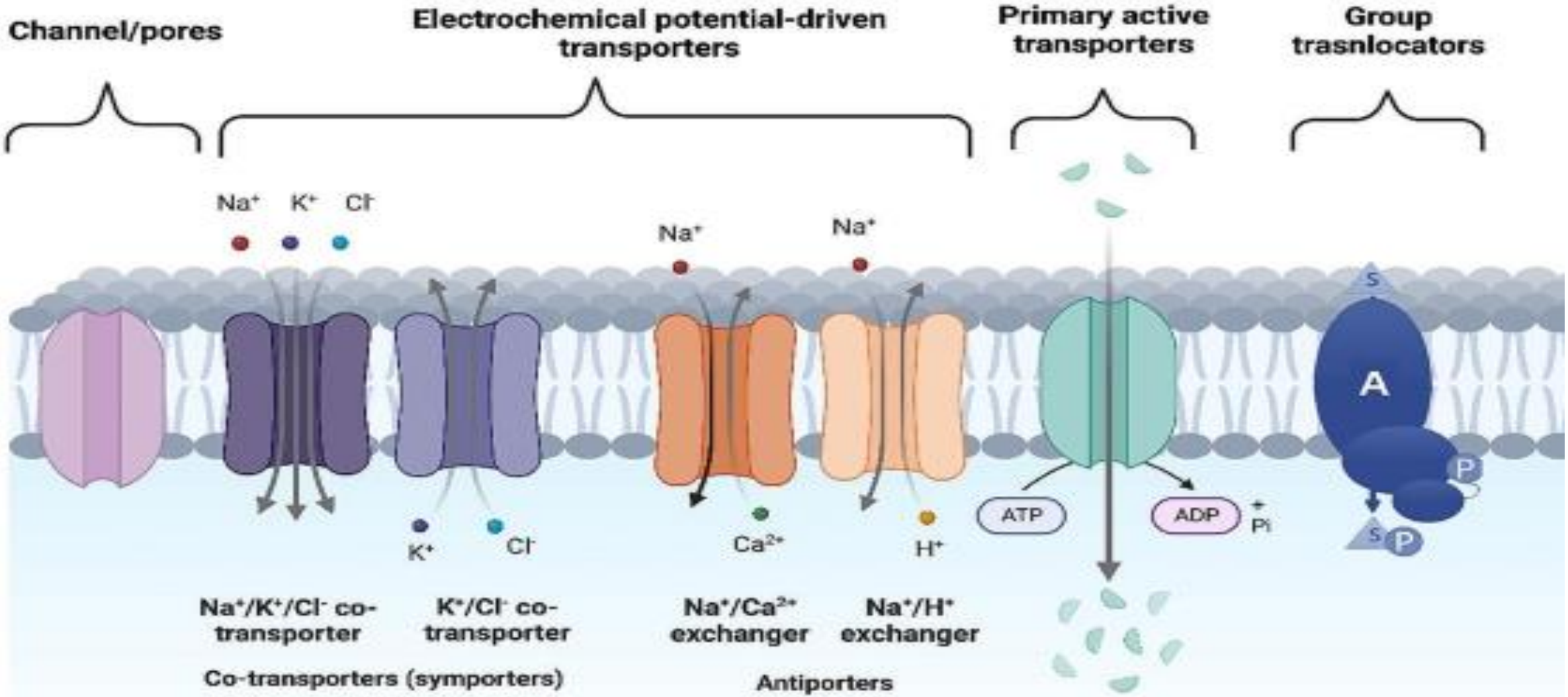
The best-known group translocation system is the *phosphoenolpyruvate: sugar phosphotransferase system (PTS)*, which transports a variety of sugars into prokaryotic cells while Simultaneously phosphorylating them using phosphoenolpyruvate (PEP) as the phosphate donor.

The phosphoenolpyruvate: sugar phosphotransferase system of *E. coli.*



Simple comparison of transport systems

Items	Passive diffusion	Facilitated diffusion	Active transport	Group translocation
carrier proteins	Non	Yes	Yes	Yes
transport speed	Slow	Rapid	Rapid	Rapid
against gradient	Non	Non	Yes	Yes
transport molecules	No specificity	Specificity	Specificity	Specificity
metabolic energy	No need	Need	Need	Need
Solutes molecules	Not changed	Changed	Changed	Changed


Channel/pores
Electrochemical potential-driven transporters
Primary active transporters
Group trasnlocators
Na+ K+ Cl-
Na+
Na+
K+
Cl-
Ca2+
H+
ATP
ADP
Pi
A
S
P
S
P
Na+/K+/Cl- co-transporter
K+/Cl- co-transporter
Na+/Ca2+ exchanger
Na+/H+ exchanger
Co-transporters (symporters)
Antiporters